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**Research Paper Analysis 1** **May 12, 2026**

## Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks - Lewis et al., 2020

RAG addresses an important limitation of parametric models: after training, they cannot update their knowledge, cannot cite sources, and hallucinate when asked about rare facts. The solution of combining BART with a DPR retriever over a Wikipedia index and training end-to-end with retrieved documents as latent variables, is a smart and promising approach.

The Jeopardy generation example shows nicely how the two memory types divide the work. When given "Hemingway" as an input, the retriever identifies the relevant books, but once BART generates the first tokens of a title like "The Sun...", the document posteriors flatten and the model completes the rest from its own parametric memory. In other words: retrieval triggers the right knowledge, parametric memory delivers the details.

The index hot-swapping experiment is the most convincing result as replacing the 2016 Wikipedia index with one from 2018 correctly updates answers about changed world leaders without any retraining, which proves the factual knowledge is contained in the index rather than the model weights.

There is also an argument for RAG in terms of efficiency since RAG-Sequence with 626M parameters outperforms T5-large with 770M on Natural Questions (44.5 vs. 28.9 exact match), because the generator does not need to memorize what it can retrieve.

A weak design choice is keeping the document encoder frozen. Only the query encoder and BART are fine-tuned, so the document representations stay fixed at DPR's initialization, which was optimized for Natural Questions and TriviaQA. As a consequence, for tasks with different retrieval characteristics the model cannot correct this.

Also the researchers never test a version where the document encoder is updated, so the claim that freezing it causes no significant performance loss is never verified. I would at minimum test a version where the document encoder is updated every N steps to see whether the practical cost is actually worth avoiding.

Beyond the encoder, the choice of Wikipedia as the sole knowledge source is also a constraint of the paper. Several questions are unanswerable from Wikipedia alone, which the authors admit, but this is not reflected in how they frame or evaluate the results.

Lastly, the claim that RAG is a general-purpose architecture is not really tested since all four tasks require retrieving short factual snippets and producing brief outputs. There is no long-form generation and no scenario where retrieved documents are actively misleading.

## PubMedQA: A Dataset for Biomedical Research Question Answering - Jin et al., 2019

PubMedQA is built on the smart observation that many PubMed articles use a research question as their title and answer it in the conclusion, which maps naturally onto a QA format. Because the question and context come from the same authors, they are matching, and the reasoning needed is quantitative since it requires comparing group statistics and interpreting p-values, in contrast to keyword matching only. That is a contribution over existing, often small biomedical QA datasets.

The main problem is class imbalance, which the paper acknowledges but does not really confront. With 55.2% "yes" labels, the majority baseline already scores 55.2% accuracy. The best model reaches 68.1% which is only 13 percentage points above always guessing. This reflects a publication bias where studies that find positive results are more likely to frame their title as a question and publish. The artificially generated PQA-A subset makes this worse, with a 92.8% yes rate, meaning Phase I pre-training exposes the model to a heavily distorted picture of biomedical evidence. The "maybe" class, which represents genuine scientific uncertainty, gets only 11% of labels and the worst model performance across all training setups. Connected to the evaluation critique in Kalai et al., under binary accuracy optimization predicting "maybe" is almost never the optimal move, meaning the evaluation structure itself discourages the behavior the dataset was designed to capture.

The paper also overstates how much quantitative reasoning is actually required. 75.5% of contexts already contain natural language interpretations of the numbers, meaning the model mostly just needs to read the text interpretation rather than interpret raw statistics.

## Why Language Models Hallucinate – Kalai et al., 2025

The paper has two main arguments.

The first, that hallucinations are statistically inevitable during pretraining, is formalized through a reduction from language generation to binary classification (Is-It-Valid). The core claim is that generating valid outputs is harder than classifying them, so any calibrated model will produce some errors. This is best illustrated by the birthday example. If a fact appears only once in the training data, the model has no way to distinguish whether it is correct or just a plausible-sounding error, so it will hallucinate on at least as many facts as there are such singletons in the training corpus.

The second argument is more important and more convincing to me. Hallucinations persist because mainstream evaluations reward them. Observation 1 shows that under any binary grader a confident wrong answer always beats "I don't know". The meta-analysis in Section 4 finds that 9 out of 10 major benchmarks give zero credit for uncertainty. The proposed fix is to add explicit confidence thresholds to existing benchmark prompts rather than introducing new hallucination-specific evals that the field would likely ignore. This is a concrete, plausible and realistic suggestion.

The paper also puts the other two papers in a sharper light. RAG is often cited as a hallucination mitigation, but Kalai et al. point out that even retrieval-augmented models face the same binary evaluation incentive. When retrieved documents are ambiguous, guessing still beats an "I don't know". And PubMedQA's "maybe" class is a direct victim of this since under accuracy optimization, expressing uncertainty is never the right move, which helps explain why no model in that paper learns to use the label reliably.